

Chapter 4 – Introduction to Problem Solving

Multiple Choice Questions (MCQs)

Introduction

1. Computers are used today mainly because they perform tasks
 - A. Slowly and manually
 - B. Faster and more accurately
 - C. Only for entertainment
 - D. Only for calculations
2. Booking train tickets online is possible mainly due to
 - A. Mobile phones
 - B. Internet cafés
 - C. Computerisation
 - D. Manual reservation
3. Railway reservation is considered a complex task because it involves
 - A. Only passenger details
 - B. Only ticket prices
 - C. Multiple trains, users, and schedules
 - D. Only payment processing
4. Online train ticket booking has increased user comfort by allowing booking
 - A. Only during office hours
 - B. Only from railway stations
 - C. Anytime and anywhere
 - D. Only through agents
5. The term *computerisation* refers to
 - A. Buying computers
 - B. Connecting computers to internet
 - C. Using computers to automate routine tasks
 - D. Learning programming languages
6. Problem solving is an essential skill especially for
 - A. Doctors
 - B. Computer science students
 - C. Shopkeepers
 - D. Drivers
7. Computers cannot solve problems on their own because they need
 - A. Electricity
 - B. Internet
 - C. Step-by-step instructions
 - D. Memory
8. The success of a computer in solving a problem depends on
 - A. Speed of the processor
 - B. Correct problem definition and solution design

- C. Cost of the computer
- D. Size of the program

9. Problem solving involves

- A. Only writing a program
 - B. Identifying problem, developing algorithm, and implementing it
 - C. Only designing output
 - D. Only testing software
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4.2 Steps for Problem Solving

10. The example of a vehicle making strange noise illustrates that

- A. All problems are easy
- B. Problems may require multiple steps to solve
- C. Machines repair themselves
- D. Noise is harmless

11. Complex problems require

- A. Guesswork
- B. A methodical approach
- C. Trial and error only
- D. Random solutions

12. Problem solving begins with

- A. Coding
- B. Testing
- C. Precise identification of the problem
- D. Debugging

13. Problem solving ends with

- A. Algorithm design
 - B. Flowchart
 - C. Working solution in the form of software
 - D. Documentation
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4.2.1 Analysing the Problem

14. Analysing a problem helps in understanding

- A. Only syntax
- B. Hardware requirements
- C. Inputs and outputs of the program
- D. Programming language

15. If a problem is not clearly understood, the developed program may

- A. Run faster
- B. Consume less memory
- C. Not solve the intended purpose
- D. Be error-free

16. During problem analysis, we identify

- A. Variables only
 - B. Core functionalities of the solution
 - C. Compiler errors
 - D. Output format only
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4.2.2 Developing an Algorithm

17. An algorithm is represented using

- A. Machine language
- B. Assembly language
- C. Natural language
- D. Binary code

18. An algorithm can be compared to

- A. A circuit diagram
- B. A well-written recipe
- C. A database
- D. A compiler

19. For a given problem

- A. Only one algorithm exists
- B. No algorithm is possible
- C. More than one algorithm may exist
- D. Algorithm is optional

20. The most suitable algorithm should be

- A. The longest one
- B. The most complex one
- C. Carefully selected
- D. Written in code

21. GIGO stands for

- A. Good Input Good Output
- B. Garbage In Garbage Out
- C. Graphical Input Graphical Output
- D. General Input General Output

22. GIGO implies that

- A. Output is independent of input
 - B. Correct input leads to wrong output
 - C. Correctness of output depends on input
 - D. Computers fix wrong inputs
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4.2.3 Coding

23. Coding is the process of

- A. Testing a program
- B. Converting algorithm into a form understood by computer

- C. Writing documentation
- D. Analysing problem

24. Programs are written using

- A. Low-level languages only
- B. High-level programming languages
- C. Binary language only
- D. Natural language

25. Documentation during coding is useful for

- A. Increasing speed
- B. Revisiting programs later
- C. Removing errors automatically
- D. Compiling code

4.2.4 Testing and Debugging

26. Testing ensures that a program

- A. Uses less memory
- B. Meets user requirements
- C. Is written quickly
- D. Has more lines of code

27. A program should generate correct output for

- A. Limited inputs
- B. Sample inputs only
- C. All possible inputs
- D. No inputs

28. Syntax errors result in

- A. Wrong output
- B. No output
- C. Partial output
- D. Slow output

29. Incorrect output despite no syntax error indicates

- A. Hardware error
- B. Logical error
- C. Network error
- D. Compiler error

30. Which of the following is NOT a standard testing method?

- A. Unit testing
- B. Integration testing
- C. Acceptance testing
- D. Random testing

31. Debugging refers to

- A. Writing code
- B. Designing algorithm

- C. Removing errors from program
- D. Running program

32. Software maintenance includes

- A. Writing algorithms
 - B. Fixing user-reported problems
 - C. Buying new computers
 - D. Creating hardware
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4.3 Algorithm

33. An algorithm is a

- A. Programming language
- B. Flowchart
- C. Finite sequence of steps
- D. Software

34. An algorithm must produce results in

- A. Infinite time
- B. Random time
- C. Finite time
- D. No time

35. An algorithm has

- A. No start or end
 - B. Only start
 - C. Only end
 - D. Definite start and end
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GCD Example

36. GCD stands for

- A. Greatest Common Difference
- B. Greatest Common Divisor
- C. General Common Divisor
- D. Global Common Division

37. The GCD of 45 and 54 is

- A. 3
- B. 6
- C. 9
- D. 18

38. Finding GCD involves

- A. Guessing numbers
- B. Listing divisors and finding the largest common one
- C. Subtraction only
- D. Multiplication

4.3.1 Why do we need an Algorithm?

39. An algorithm acts as a

- A. Compiler
- B. Roadmap for program
- C. Debugger
- D. Database

40. Without an algorithm, a programmer may

- A. Write efficient code
- B. Clearly visualise solution
- C. Develop a faulty program
- D. Increase performance

41. Algorithms are used in

- A. Search engines
- B. Online banking
- C. Booking taxis
- D. All of the above

42. Writing an algorithm is considered

- A. Last step of programming
- B. First step of programming
- C. Optional step
- D. Debugging step

43. A correct algorithm ensures

- A. Computer never crashes
- B. Program runs correctly every time
- C. Faster internet
- D. Better hardware

Characteristics of a Good Algorithm

44. Precision in an algorithm means

- A. Steps are vague
- B. Steps are precisely defined
- C. Steps depend on guess
- D. Steps are optional

45. Uniqueness ensures that

- A. Steps repeat
- B. Output depends on previous results and input
- C. Output is random
- D. Steps are ignored

46. Finiteness means the algorithm

- A. Runs forever
- B. Stops after finite steps

- C. Never ends
 - D. Skips steps
- 47.** An algorithm must have
- A. Input only
 - B. Output only
 - C. Both input and output
 - D. Neither input nor output
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While Writing an Algorithm

- 48.** While writing an algorithm, one must clearly identify
- A. Color of interface
 - B. Input, processing, and output
 - C. Hardware brand
 - D. Internet speed
- 49.** Processing refers to
- A. Input collection
 - B. Computation to get result
 - C. Output display
 - D. Program testing
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Pseudocode and Flowchart

- 50.** Pseudocode is mainly used to
- A. Execute programs faster
 - B. Represent the functionality of a program in human-readable form
 - C. Replace flowcharts
 - D. Reduce memory usage
- 51.** Benefits of writing pseudocode before coding include
- A. Ensuring no step is left out
 - B. Making code unreadable
 - C. Reducing computer speed
 - D. Avoiding algorithm creation
- 52.** In the example of calculating area and perimeter of a rectangle, the output includes
- A. Only Area
 - B. Only Perimeter
 - C. Both Area and Perimeter
 - D. Length and Breadth
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Flow of Control

- 53.** Flow of control in an algorithm refers to
- A. Memory usage of a program
 - B. Execution order of statements

- C. Speed of a processor
- D. Cost of coding

54. If all steps in an algorithm execute one after the other, it is called

- A. Repetition
- B. Selection
- C. Sequence
- D. Conditional

55. Which control structure is used when a step needs to be repeated a known number of times?

- A. Sequence
 - B. Selection
 - C. Loop / Iteration
 - D. Conditional
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Selection / Decision Making

56. A decision-making algorithm is used when

- A. All steps are fixed
- B. Output depends on a condition
- C. No computation is needed
- D. There are no inputs

57. True/False values in a conditional are also called

- A. Decimal values
- B. Binary values
- C. ASCII codes
- D. Random values

58. In programming, the keyword representing the “otherwise” part of a condition is

- A. If
- B. Else
- C. While
- D. Repeat

59. For checking whether a student is a child (<13), teenager (≥ 13 but <20), or adult (≥ 20), which control structure is used?

- A. Sequence only
- B. Selection / If-Else
- C. Loop / Iteration
- D. None of the above

60. In the “Dragons and Wizards” card game, a diamond or club card will

- A. Give a point to Wizards
- B. Give a point to Dragons
- C. End the game
- D. Reset points

61. If a heart card is drawn and its value is a number, which team gets a point?

- A. Dragons
 - B. Wizards
 - C. Both
 - D. None
-

Repetition / Loops

62. Repetition in an algorithm is also called

- A. Sequence
- B. Iteration / Loop
- C. Selection
- D. Conditional

63. Example 4.8 (Average of 5 numbers) uses which type of loop?

- A. WHILE loop
- B. FOR loop (fixed number of iterations)
- C. If-Else
- D. Sequence

64. In the example of calculating average until user enters 0, the number of repetitions is

- A. Known before starting
- B. Unknown before starting
- C. Always 5
- D. Always 10

65. In repetition structures, a **counter variable** is used to

- A. Store input numbers
- B. Track the number of times loop has executed
- C. Compute output only
- D. Print results

66. In example 4.9, why is the INPUT statement used twice?

- A. To confuse the user
- B. To initialize and then update the input for checking the loop condition
- C. To store sum
- D. To calculate average

67. In calculating the average of numbers until 0 is entered, the average is calculated by

- A. sum / 5
- B. sum / count
- C. sum * count
- D. sum + count

68. Looping until a condition becomes false is an example of

- A. Fixed iteration
- B. Conditional iteration
- C. Sequence execution
- D. Selection

69. Which flowchart symbol is mainly used to depict decision-making?

- A. Rectangle
- B. Diamond
- C. Oval
- D. Parallelogram

70. In the pseudocode for calculating area and perimeter, the **COMPUTE** keyword indicates

- A. Taking input
- B. Performing calculation
- C. Printing output
- D. Ending the program

71. Which of the following is NOT an example of repetition in daily life?

- A. Clapping 5 times
- B. Walking 10 steps ahead
- C. Jumping until tired
- D. Choosing a route based on traffic

72. What type of control structure is used in the card game algorithm for multiple conditional checks?

- A. Sequence only
- B. Nested If-Else
- C. Loop only
- D. None

Algorithm Verification

73. Why is it important to verify an algorithm before developing software?

- A. To make the software colorful
- B. To ensure it works correctly for all input values
- C. To reduce the size of the program
- D. To make programming easier

74. A dry run of an algorithm involves

- A. Writing code in a high-level language
- B. Manually going through the steps with different input values
- C. Printing the output only
- D. Running the program on the computer

75. In the example of adding times $T1 = 4 \text{ hrs } 50 \text{ mins}$ and $T2 = 2 \text{ hrs } 20 \text{ mins}$, why was the algorithm modified?

- A. To use fewer variables
- B. To adjust minutes greater than 60
- C. To decrease hours
- D. To simplify input

76. Verification of algorithms is important because

- A. Errors are easier and cheaper to fix at this stage
- B. It increases the program size
- C. It reduces memory requirements
- D. It makes compilation faster

Comparison of Algorithms

77. Which of the following is NOT a factor used to compare algorithms?

- A. Time complexity
- B. Space complexity
- C. Color of the code
- D. Efficiency

78. Algorithm (iii) for checking prime numbers is more efficient than (i) and (ii) because it

- A. Uses only odd numbers
- B. Checks divisibility only up to the square root of the number
- C. Uses a pre-stored list of primes
- D. Does not check any divisibility

79. Algorithm (iv) for checking prime numbers is efficient because

- A. It requires no memory
- B. It checks all numbers till the dividend
- C. It uses only pre-stored prime numbers for divisibility
- D. It ignores numbers greater than 100

80. Time complexity refers to

- A. The memory required by an algorithm
- B. The execution time required by an algorithm
- C. The number of programmers needed
- D. The type of input used

81. Space complexity refers to

- A. Amount of memory required to execute an algorithm
- B. Time taken to execute an algorithm
- C. Number of lines in the program
- D. Number of functions used

Coding and High-Level Languages

82. Programs written in high-level languages are easier to

- A. Read, write, and maintain
- B. Understand by the computer directly
- C. Convert into low-level machine code manually
- D. Run without compilation

83. Which of the following is NOT a high-level programming language?

- A. Python
- B. Java
- C. C++
- D. Machine language

84. Source code is

- A. The compiled machine code
- B. The program written in a high-level language

- C. A list of algorithms only
- D. The memory of the computer

85. Why are high-level languages called portable?

- A. They do not require a compiler
 - B. They can run on different types of computers with little or no modification
 - C. They work only on desktops
 - D. They run without a source code
-

Decomposition

86. What is the main idea behind decomposition in problem solving?

- A. Solve problems using only one algorithm
- B. Break a complex problem into smaller sub-problems
- C. Avoid testing the program
- D. Run the program faster

87. In decomposition, solving sub-problems independently is beneficial because

- A. It reduces overall complexity
- B. It avoids writing algorithms
- C. It eliminates the need for testing
- D. It removes all input values

88. Which of the following is an example of decomposition in real life?

- A. Solving a complex math problem by dividing it into smaller steps
- B. Writing all code in one single block
- C. Running a program without verification
- D. Ignoring user inputs

89. After solving individual sub-problems in decomposition, the next step is

- A. Ignore testing and integration
- B. Combine them logically to get the complete solution
- C. Delete the sub-problems
- D. Write new unrelated programs

90. In the railway reservation system example, decomposition helps to

- A. Reduce ticket prices
- B. Assign different components like train info, booking, staff management to different teams
- C. Avoid using computers
- D. Increase the number of stations